

NELIREF Python Programming - Task 1 Assignment

Building Discipline in AI-Assisted Learning with Vibe Coding

Course: NELIREF 3rd Data Science & AI Summer School 2026

Assignment: Python Programming - Task 1

Difficulty Level: Beginner

Due Date: Sunday, June 28, 2026 at 11:59 PM Eastern Daylight Time (EDT)

Submission: Email your repository link to **newliferesearchconsult@gmail.com**

Assignment Overview

This assignment is designed to instill **discipline** and **best practices** in your programming journey. You will:

- Learn to use **GitHub** for version control
- Master **GitHub Codespaces** for cloud-based development
- Practice **Vibe Coding** - learning with AI assistance
- Run and debug actual Python notebooks
- Complete practical coding exercises
- Document your learning journey with git commits

Key Philosophy: This is not about getting perfect code - it's about **building habits, understanding concepts**, and learning to **effectively use modern AI tools** to accelerate your learning.

Learning Objectives

By completing this assignment, you will:

1. **Develop GitHub fluency** - Fork repositories, create branches, commit changes, and push to main
 2. **Master AI-assisted coding** - Use Copilot, Claude, or ChatGPT effectively to understand and fix code
 3. **Practice Python fundamentals** - Understand data types, operations, and control flow
 4. **Build documentation habits** - Comment code changes and document your learning
 5. **Create reproducible workflows** - Ensure all work is saved and version-controlled
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Prerequisites

Before starting, ensure you have:

- A **GitHub account** (free at github.com)
- Basic understanding of **git** (clone, commit, push)
- Access to **GitHub Copilot** (free for everyone, education benefits for students and staff of educational institutions with institutional emails, or use free alternatives like Claude/ChatGPT)
- A **Python environment** (Codespace, local machine, or cloud IDE such as Google Colab, Anaconda, etc)
- Familiarity with **Jupyter Notebooks**

Don't have these? Don't have these? Check the Resources section at the end for setup guides or visit the Python Programming repository for this assignment to download materials discussed during the instruction days.

Task Breakdown

PART 1: Repository Setup (Due: Sunday, June 28)

Step 1a: Fork or Clone the Repository

Option A: Fork the Repository (Recommended for Learning)

*# Go to <https://github.com/newliferesearchfoundation/python-programming>
Click the "Fork" button in the top-right
This creates YOUR OWN copy of the repository*

Option B: Clone the Repository

```
git clone https://github.com/newliferesearchfoundation/python-programming.git
cd python-programming
git remote add origin https://github.com/YOUR-USERNAME/python-programming.git
git push -u origin main
```

Option C: Download & Create New Repository

*# Download the ZIP file
Extract it
Create a new repository on GitHub
Upload the content
git clone your-new-repo*

Step 1b: Create Your GitHub Codespace

1. **Go to your forked repository** on GitHub
2. **Click "Code"** button (green button, top-right)
3. **Select "Codespaces"** tab

4. Click **“Create codespace on main”**
 5. Wait for Codespace to initialize (2-3 minutes)
 6. You now have a cloud-based VS Code environment!
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PART 2: Set Up AI Assistance (Due: Sunday, June 28)

GitHub Copilot Setup (Recommended)

1. In your Codespace, open the **Extensions** marketplace
2. Search for **“GitHub Copilot”**
3. Click **“Install”**
4. **Sign in** with your GitHub account
5. You’re ready to use Copilot!

Alternative AI Tools (macOS Users or Preference)

- **Claude:** Use in VS Code via Claude extension
- **ChatGPT:** Use in a browser tab alongside your editor
- **ChatGPT-4o:** Best for detailed explanations

Pro Tip: You don’t need to be “perfect” at AI usage yet. Use it to: - Understand error messages - Explain code - Fix broken cells - Verify your solutions

PART 3: Run & Debug Notebooks (Due: Sunday, June 28)

Task 3a: Run 2026_Python_Programming_Notebook.ipynb

1. **Navigate to the notebook** in your Codespace and don’t forget to select the Kernel
2. **Install Jupyter** if needed:

```
pip install jupyter notebook
```
3. **Start Jupyter:**

```
jupyter notebook
```
4. **Open the notebook** 2026_Python_Programming_Notebook.ipynb
5. **Run each cell** (Shift + Enter) and study them, ensure you understand what’s happening in each cell
6. **Fix any broken cells:**
 - Read the error message
 - Use Copilot to suggest fixes: // Ask Copilot: Why is this cell breaking?

- **Document the fix** with a comment in the cell
- Example comment:

```
# Fixed by Copilot: Changed variable name from 'x' to 'num'
# to avoid conflict with imported variable
num = 42
print(num)
```

Task 3b: Run `Python_Basics_for_Beginners.ipynb`

1. **Open the notebook** in your Codespace
2. **Read each section carefully** and run the examples
3. **Complete Section 7: Practice Exercises**
 - Exercise 1: Temperature Converter
 - Exercise 2: Sum Calculator
 - Exercise 3: Letter Counter
 - Exercise 4: Even Number Filter
 - Exercise 5: Rock-Paper-Scissors Game

How to Approach These:

1. **Read the exercise description** first
2. **Try solving it yourself** (5-10 minutes)
3. **If stuck, use Copilot/AI:** - Ask: "How can I approach this?" - Ask: "Show me a similar example" - Ask: "What's wrong with my code?"
4. **Understand the solution** – Plan with Copilot by using the Plan mode on Copilot before using the Agent mode to run the notebook
5. **Add a comment** explaining what you learned:


```
python      # Learned: Using
isinstance() to check data types      # Copilot helped me understand the
conditional logic      def get_even_numbers(numbers):      return [n for n in
numbers if n % 2 == 0]
```

PART 4: BONUS Challenge (Optional, Due: Sunday, June 28)

Create a game in a separate notebook!

Requirements: - Create a new notebook: `my_game.ipynb` - Build ANY game using Python basics you learned - Use Copilot to help with planning, logic and structure - Document your code with comments by using Copilot's Agent mode for execution

Game Ideas: - Hangman (guess the word) - Guess the Number (higher/lower game) - Quiz Game (multiple choice questions) - Card Game (simple card logic) - Dice Rolling Game - Choose Your Own Adventure (text-based)

Example Structure:

```
# Game: Guess the Number  
# Created with Copilot assistance  
# Learned: While loops, input validation, random module
```

```
import random  
  
def play_game():  
    """Main game function"""  
    secret_number = random.randint(1, 100)  
    guesses = 0  
  
    while True:  
        guess = int(input("Guess a number (1-100): "))  
        guesses += 1  
  
        if guess < secret_number:  
            print("Too low!")  
        elif guess > secret_number:  
            print("Too high!")  
        else:  
            print(f"You won in {guesses} guesses!")  
            break  
  
play_game()
```

PART 5: Version Control & Submission

Step 5a: Commit Your Changes

In your Codespace terminal, run:

```
# 1. Check what files changed  
git status
```

```
# 2. Add all changes  
git add .
```

```
# 3. Commit with a meaningful message  
git commit -m "Task 1: Completed Python basics exercises and fixed notebooks"
```

- Fixed broken cells in 2026_Python_Programming_Notebook.ipynb
- Completed all 5 practice exercises in Python_Basics_for_Beginners.ipynb
- Added comments documenting Copilot-assisted learning
- [Optional] Created bonus game: my_game.ipynb"

4. Push to GitHub
git push origin main

Step 5b: Verify Your Changes

1. **Go to your GitHub repository**
2. **Refresh the page** to see your changes
3. **Click on commits** to verify your commit message
4. **Open the notebooks** to confirm changes are saved

Step 5c: Submit Your Work

Email: newliferesearchconsult@gmail.com

Subject Line:

NELIREF Task 1 Submission - [YOUR NAME] - Python Programming

Email Body:

Hello,

Please find my Task 1 submission below:

GitHub Repository: <https://github.com/YOUR-USERNAME/python-programming>

Completed:

1. Forked/cloned python-programming repository
2. Created GitHub Codespace
3. Set up Copilot (or alternative AI tool)
4. Ran and fixed 2026_Python_Programming_Notebook.ipynb
5. Completed all exercises in Python_Basics_for_Beginners.ipynb
6. [Optional] Created bonus game
7. Committed and pushed all changes to main branch

AI Tools Used: [Copilot / Claude / ChatGPT / ChatGPT-4o]

Platform: [GitHub Codespace / VS Code Local / macOS / Other]

Thank you,
[YOUR NAME]

Grading Rubric

Criteria	Points	Requirements
Repository Setup	15	Fork/clone complete, Codespace created, accessible
Notebook Execution	20	Both notebooks run successfully, broken cells fixed
Practice Exercises	40	All 5 exercises completed correctly with comments
Code Documentation	15	Clear comments explaining changes and learning

Criteria	Points	Requirements
Git Workflow	10	Proper commits, meaningful messages, pushed to main
Bonus Game	20	Optional: +20 bonus points for creative game
TOTAL	100	(Plus 20 bonus)

Troubleshooting Common Issues

“Copilot is not working in my Codespace”

- Make sure you’re signed into GitHub
- Restart Codespace
- Try the GitHub Copilot extension

“My notebook has a kernel error”

In terminal:

```
pip install --upgrade jupyter ipykernel
python -m ipykernel install --user
```

“Git push is being rejected”

Make sure you're on main branch
`git branch`

If not on main:

```
git checkout main
git pull origin main
git push origin main
```

“I can’t access Copilot (not a student/not free)”

- Codespaces is free for everyone
- Use **Claude** or **ChatGPT** instead if codespaces fail (both have free plans)
- Ask in the same way: “Fix this error” or “Explain this code”
- The learning objective is the same

“I don’t understand an error message”

- Copy the ENTIRE error message
 - Paste it to Copilot/Claude/ChatGPT with: “What does this error mean?”
 - Read the explanation
 - Try the suggested fix
 - Ask NELIREF Summer School Instructors and TA on Telegram at <https://t.me/neliref>
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Resources & Support

GitHub & Git Guides

- [GitHub Hello World \(5 min\)](#)
- [GitHub Codespaces Documentation](#)
- [Git Basics \(Official Git Book\)](#)

Python Resources

- [Python Official Documentation](#)
- [Automate the Boring Stuff with Python \(Free book\)](#)
- [Real Python Tutorials](#)

AI Assistance & Vibe Coding

- [GitHub Copilot Documentation](#)
- [Using Claude for Code](#) - Free tier available
- [ChatGPT for Beginners](#)
- **Vibe Coding Philosophy:** Use AI as a learning partner, you remain the boss.

Jupyter Notebook Help

- [Jupyter Official Docs](#)
- [Notebooks Tutorial](#)

Getting Help

- **Ask Copilot/Claude:** "I'm getting X error, how do I fix it?"
- **Check Stack Overflow:** Search your error message
- **NELIREF Social media/Chat:** Ask community members www.neliref.org to join social media platforms
- **Email instructors:** newliferesearchconsult@gmail.com

Tips for Success

Best Practices

1. **Code Daily** - Even 30 minutes a day beats 6 hours once a week
2. **Understand Before Copying or allowing code** - Read Copilot's suggestions and code
3. **Comment Everything** - Future you will thank present you, Copilot can help with this as well
4. **Use Git Commits** - Commit every time you finish a section, not just at the end
5. **Test Your Code** - Run cells multiple times to make sure they work

Using AI Assistants Effectively

DO: - Ask **specific questions**: “Why is this line breaking?” - Request **explanations**: “Explain what this function does” - Ask for **help understanding**: “What does a list comprehension do?” - Request **similar examples**: “Show me how to use `isinstance()`”

DON'T: - Just copy-paste code or use Copilot in writing a notebook without reading it - Skip understanding the explanation is not helpful - Use AI as a replacement for thinking, use it to plan together

Building the Discipline

The goal of this assignment is to help you: - **Think like a programmer** - solving problems systematically - **Learn from errors** - understanding what went wrong and why - **Use modern tools** - AI is standard in 2026 programming - **Document your work** - writing comments and commit messages - **Version your code** - saving your progress safely

Success Looks Like

By Sunday, June 28 at 11:59 PM EDT, you should have:

- **A GitHub repository** with your forked/cloned code
 - **Multiple git commits** showing your progress
 - **Both notebooks executed** without errors
 - **All 5 exercises completed** in `Python_Basics_for_Beginners.ipynb`
 - **Comments in your code** explaining what you learned
 - **[Bonus] A game notebook** (optional)
 - **Your repository link** submitted via email
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Frequently Asked Questions

Q: Can I use a different AI tool instead of Copilot?

A: Yes! Claude, ChatGPT, and ChatGPT-4o work great. The skill being practiced is the same.

Q: What if I don't understand the notebook content?

A: Use AI to help explain it. Ask: “Explain this Python code in simple terms.” Then run it step-by-step.

Q: Do I have to complete the bonus game?

A: No, it's optional. But it's a great way to practice and have fun!

Q: What if I can't finish by Sunday?

A: Submit what you have. Partial work is better than no submission. Email if you need accommodation. **Changes pushed to this repository after the deadline will also be accepted until July 28, 2026.**

Q: Can I work with a partner?

A: You can discuss concepts, but each person must complete and submit their own repository.

Q: How detailed should my comments be?

A: At least one comment per exercise explaining what the code does or what you learned.

Q: What if my code doesn't match the expected output exactly?

A: Focus on understanding why. Email the instructors and teaching assistants if you're stuck.

Support Contact

For Questions or Issues: - Email: newliferesearchconsult@gmail.com - Subject: "NELIREF Task 1 - [Your Question]"

For Technology Help: - GitHub Copilot Issues: Check GitHub Copilot status page - Codespace Issues: See GitHub status page - Python/Notebook Issues: Post to Stack Overflow with error message

Timeline Summary

Date	Milestone	Status
June 21, 2026 (EDT)	Assignment Released	Today
June 22-28, 2026	Work on Tasks	In Progress
June 28, 2026 (11:59 PM EDT)	DEADLINE	Due
July 28, 2026	Feedback Provided	After deadline

Final Thoughts

This assignment is about **building excellence through discipline**. The skills you're learning - version control, AI-assisted coding, documentation are the same skills used by professional software engineers at Google, Microsoft, Meta, and everywhere else.

You're not just learning Python. You're learning **how to be a programmer in 2026**.

Let's get started!

Happy Coding!

Created for NELIREF 3rd Data Science & AI Summer School 2026

Emphasizing Discipline, Vibe Coding, and Modern Development Practices